Real Analysis Review

Sequences, Continuity, and Convergence

Phase 0: Prerequisites Review | Document 1 of 3
Primary texts: Rudin, *Principles of Mathematical Analysis*; Abbott, *Understanding Analysis*

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

February 2026

Abstract

This document is a self-contained refresher of undergraduate real analysis, aimed at readers who have seen this material before but need the key results recalled with full formal statements and complete proofs. The selection of topics is driven entirely by what is needed for measure theory, probability, and functional analysis in Phase 01 and beyond: the completeness of $\mathbb{R}$, convergence of sequences and series, topology of the real line, continuity, differentiation, Riemann integration, sequences of functions, and metric spaces. Every theorem is proved. The reader who works through this review will be fully prepared for the construction of σ-algebras, measures, and the Lebesgue integral.

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1 Introduction

This document is not a first course in real analysis. It is a focused refresher for readers who have taken undergraduate analysis but find the details "dusty." Our goal is to recall — with full proofs — exactly the results that will be needed when we move to measure theory and beyond.

Here is what we cover and why each topic matters:

1. **The real number system** (Section 2). The completeness axiom (least upper bound property) is the single most important property of $\mathbb{R}$. It drives the construction of Lebesgue measure, ensures the existence of suprema needed to define outer measures, and underlies every convergence theorem in the curriculum.
2. **Sequences and series** (Section 3). The monotone convergence theorem for sequences, Bolzano–Weierstrass, Cauchy completeness, and $\limsup / \liminf$ are the workhorses of measure-theoretic arguments. The $\limsup$ of a sequence of sets, for instance, is built directly from the $\limsup$ concept reviewed here.
3. **Topology of the real line** (Section 4). Open sets, closed sets, and compactness are the language in which σ-algebras and Borel sets are defined. The Heine–Borel theorem characterizes compact subsets of $\mathbb{R}^n$.
4. **Continuity** (Section 5). Continuous functions on compact sets attain their bounds and are uniformly continuous — facts used repeatedly in measure theory and functional analysis.
5. **Differentiation** (Section 6). The mean value theorem and Taylor's theorem are essential for optimization theory and appear in many ε-δ arguments throughout the curriculum.
6. **Riemann integration** (Section 7). Understanding the Riemann integral — and its limitations — motivates the passage to the Lebesgue integral in Phase 01.
7. **Sequences and series of functions** (Section 8). Uniform convergence and its interchange theorems are the starting point; the dominated and monotone convergence theorems of measure theory vastly generalize these results.
8. **Metric spaces** (Section 9). The abstract setting of metric spaces is where functional analysis begins. The contraction mapping theorem proved here is directly used to establish convergence of value iteration in reinforcement learning.

From Real Analysis to Measure Theory

Measure theory generalizes Riemann integration to handle limits of functions that Riemann cannot. The concepts in this review — $\sup / \inf$, sequences, limits, open/closed sets, compactness — are the raw material from which σ-algebras, measures, and Lebesgue integration are built.

2 The Real Number System

2.1 Ordered Field Axioms

We take the real numbers $\mathbb{R}$ as given, satisfying the following axioms.

Definition 2.1 (Ordered Field)**.** The real numbers $\mathbb{R}$ form an *ordered field*: a set equipped with operations $+$ and $\cdot$ and a total order $\leq$ satisfying:

(i) **Field axioms.** $(\mathbb{R}, +, \cdot)$ is a field: addition and multiplication are commutative, associative, and distributive, with identity elements 0 and 1 respectively, additive inverses for all elements, and multiplicative inverses for all nonzero elements.

(ii) **Order axioms.** $\leq$ is a total order compatible with the field operations: if $a \leq b$ then $a + c \leq b + c$ for all c; if $0 \leq a$ and $0 \leq b$ then $0 \leq ab$.

Remark 2.1. The rationals $\mathbb{Q}$ also form an ordered field. What distinguishes $\mathbb{R}$ from $\mathbb{Q}$ is the completeness axiom below.

2.2 The Completeness Axiom

Definition 2.2 (Upper Bound, Supremum)**.** Let $S \subseteq \mathbb{R}$ be nonempty.

(i) An element $u \in \mathbb{R}$ is an *upper bound* for S if $s \leq u$ for every $s \in S$. We say S is *bounded above* if such a u exists.

(ii) An element $\alpha \in \mathbb{R}$ is the *supremum* (or *least upper bound*) of S, written $\alpha = \sup S$, if:
   (a) α is an upper bound for S, and
   (b) if u is any upper bound for S, then $\alpha \leq u$.

Definition 2.3 (Lower Bound, Infimum)**.** Dually, $\ell \in \mathbb{R}$ is a *lower bound* for S if $\ell \leq s$ for every $s \in S$. The *infimum* $\inf S$ is the greatest lower bound.

Remark 2.2. The supremum $\alpha = \sup S$ is characterized by: α is an upper bound for S, and for every $\varepsilon > 0$ there exists $s \in S$ with $s > \alpha - \varepsilon$. This characterization is used constantly.

Theorem 2.1 (Completeness Axiom / Least Upper Bound Property)**.** *Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.*

This is an *axiom*: it is not proved from more basic principles, but rather is the defining property that distinguishes $\mathbb{R}$ from $\mathbb{Q}$.

Corollary 2.2 (Greatest Lower Bound Property)**.** *Every nonempty subset of $\mathbb{R}$ that is bounded below has an infimum in $\mathbb{R}$.*

Proof. Let $S \subseteq \mathbb{R}$ be nonempty and bounded below. Define $-S = \{-s : s \in S\}$. Then $-S$ is nonempty and bounded above (if ℓ is a lower bound for S, then $-\ell$ is an upper bound for $-S$). By the completeness axiom, $\alpha = \sup(-S)$ exists. We claim $\inf S = -\alpha$.

Since α is an upper bound for $-S$, we have $-s \leq \alpha$ for all $s \in S$, hence $s \geq -\alpha$ for all $s \in S$, so $-\alpha$ is a lower bound for S. If ℓ is any lower bound for S, then $-\ell$ is an upper bound for $-S$, so $\alpha \leq -\ell$, hence $\ell \leq -\alpha$. Thus $-\alpha$ is the greatest lower bound. □

2.3 Archimedean Property and Density of Rationals

Theorem 2.3 (Archimedean Property)**.** *For every $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ with $n > x$. Equivalently, for every $\varepsilon > 0$ and every $M > 0$, there exists $n \in \mathbb{N}$ with $n\varepsilon > M$.*

Proof. Suppose for contradiction that $\mathbb{N}$ is bounded above in $\mathbb{R}$. By the completeness axiom, $\alpha = \sup \mathbb{N}$ exists. Then $\alpha - 1$ is not an upper bound for $\mathbb{N}$ (since α is the *least* upper bound), so there exists $m \in \mathbb{N}$ with $m > \alpha - 1$. But then $m + 1 > \alpha$ with $m + 1 \in \mathbb{N}$, contradicting the fact that α is an upper bound for $\mathbb{N}$.

The second formulation follows by applying the first to $x = M/\varepsilon$. □

Theorem 2.4 (Density of Rationals)**.** *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $q \in \mathbb{Q}$ with $a < q < b$.*

Proof. By the Archimedean property, choose $n \in \mathbb{N}$ with $n(b - a) > 1$, i.e., $nb - na > 1$. Again by the Archimedean property, the sets $\{k \in \mathbb{Z} : k > na\}$ and $\{k \in \mathbb{Z} : k < na\}$ are both nonempty, so we may define $m = \min\{k \in \mathbb{Z} : k > na\}$ (this minimum exists because the positive integers exceeding na form a nonempty subset of $\mathbb{N}$, which has a least element by the well-ordering principle). Then $m > na$ and $m - 1 \leq na$, so $m \leq na + 1 < nb$ (since $nb - na > 1$). Setting $q = m/n$ gives $a < q < b$ with $q \in \mathbb{Q}$. □

Theorem 2.5 (Nested Intervals Property)**.** *Let $I_n = [a_n, b_n]$ be a sequence of closed bounded intervals with $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$. Then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.*

Proof. The nesting condition gives $a_1 \leq a_2 \leq \cdots$ and $b_1 \geq b_2 \geq \cdots$ with $a_n \leq b_n$ for all n. In particular, $a_n \leq b_1$ for all n, so the set $\{a_n : n \in \mathbb{N}\}$ is bounded above. By the completeness axiom, $\alpha = \sup\{a_n : n \in \mathbb{N}\}$ exists.

For every n, we have $a_n \leq \alpha$ (since α is an upper bound) and $\alpha \leq b_n$ (since b_n is an upper bound for $\{a_k\}$: for $k \leq n$, $a_k \leq a_n \leq b_n$; for $k > n$, $a_k \leq b_k \leq b_n$). Therefore $\alpha \in [a_n, b_n]$ for every n, giving $\alpha \in \bigcap_{n=1}^{\infty} I_n$. □

Remark 2.3. The conclusion fails for open intervals: $\bigcap_{n=1}^{\infty}(0, 1/n) = \emptyset$. It also fails for unbounded closed sets: $\bigcap_{n=1}^{\infty}[n, \infty) = \emptyset$. Both closure and boundedness are essential.

3 Sequences and Series

3.1 Convergence of Sequences

Definition 3.1 (Convergence)**.** A sequence $(a_n)_{n=1}^{\infty}$ in $\mathbb{R}$ *converges* to $L \in \mathbb{R}$, written $\lim_{n\to\infty} a_n = L$ or $a_n \to L$, if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for all $n \geq N$. A sequence that does not converge is *divergent*.

Theorem 3.1 (Uniqueness of Limits)**.** *If $a_n \to L$ and $a_n \to M$, then $L = M$.*

Proof. Let $\varepsilon > 0$. There exist $N_1, N_2 \in \mathbb{N}$ such that $|a_n - L| < \varepsilon/2$ for $n \geq N_1$ and $|a_n - M| < \varepsilon/2$ for $n \geq N_2$. For $n \geq \max(N_1, N_2)$:

$$|L - M| \leq |L - a_n| + |a_n - M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $|L - M| < \varepsilon$ for every $\varepsilon > 0$, we conclude $L = M$. □

Theorem 3.2 (Algebraic Limit Theorems)**.** *Suppose $a_n \to a$ and $b_n \to b$. Then:*

(i) $a_n + b_n \to a + b$.
(ii) $a_n \cdot b_n \to a \cdot b$.
(iii) If $b \neq 0$, then $a_n/b_n \to a/b$ (for n large enough that $b_n \neq 0$).

Proof. We prove (ii); the proofs of (i) and (iii) are similar.

Since (a_n) converges, it is bounded: there exists $M > 0$ with $|a_n| \leq M$ for all n. Let $\varepsilon > 0$. Choose N_1 so that $|a_n - a| < \varepsilon/(2(|b| + 1))$ for $n \geq N_1$, and N_2 so that $|b_n - b| < \varepsilon/(2M)$ for $n \geq N_2$. For $n \geq \max(N_1, N_2)$:

$$\begin{aligned} |a_n b_n - ab| &= |a_n b_n - a_n b + a_n b - ab| \\ &\leq |a_n||b_n - b| + |b||a_n - a| \\ &\leq M \cdot \frac{\varepsilon}{2M} + |b| \cdot \frac{\varepsilon}{2(|b| + 1)} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

□

3.2 Monotone Convergence and Bolzano–Weierstrass

Theorem 3.3 (Monotone Convergence Theorem for Sequences)**.** *Every bounded monotone sequence converges. Specifically:*

(i) If (a_n) is increasing and bounded above, then $a_n \to \sup\{a_n : n \in \mathbb{N}\}$.
(ii) If (a_n) is decreasing and bounded below, then $a_n \to \inf\{a_n : n \in \mathbb{N}\}$.

Proof. We prove (i). Let $\alpha = \sup\{a_n : n \in \mathbb{N}\}$, which exists by the completeness axiom. Let $\varepsilon > 0$. Since $\alpha - \varepsilon$ is not an upper bound for $\{a_n\}$, there exists N with $a_N > \alpha - \varepsilon$. Since (a_n) is increasing, for $n \geq N$:

$$\alpha - \varepsilon < a_N \leq a_n \leq \alpha,$$

so $|a_n - \alpha| < \varepsilon$. Part (ii) follows by applying (i) to $(-a_n)$. □

Theorem 3.4 (Bolzano–Weierstrass)**.** *Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.*

Proof. Let (a_n) be bounded. We construct a monotone subsequence, which converges by Theorem 3.3.

Call index n a *peak* if $a_n \geq a_m$ for all $m > n$. There are two cases:

Case 1: There are infinitely many peaks $n_1 < n_2 < n_3 < \cdots$. Then $a_{n_1} \geq a_{n_2} \geq a_{n_3} \geq \cdots$ is a decreasing subsequence. It is bounded (since (a_n) is bounded), hence convergent by Theorem 3.3.

Case 2: There are only finitely many peaks. Let N be larger than all peak indices. Then no index $n \geq N$ is a peak, meaning for every $n \geq N$ there exists $m > n$ with $a_m > a_n$. Starting with $n_1 = N$, inductively choose $n_{k+1} > n_k$ with $a_{n_{k+1}} > a_{n_k}$. This gives a strictly increasing subsequence. It is bounded above (since (a_n) is bounded), hence convergent by Theorem 3.3. □

3.3 Cauchy Sequences and Completeness

Definition 3.2 (Cauchy Sequence)**.** A sequence (a_n) in $\mathbb{R}$ is *Cauchy* if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - a_m| < \varepsilon$ for all $n, m \geq N$.

Theorem 3.5 (Completeness of $\mathbb{R}$)**.** *A sequence in $\mathbb{R}$ converges if and only if it is Cauchy.*

Proof. ($\Rightarrow$) Suppose $a_n \to L$. Given $\varepsilon > 0$, choose N so that $|a_n - L| < \varepsilon/2$ for $n \geq N$. For $n, m \geq N$:

$$|a_n - a_m| \leq |a_n - L| + |L - a_m| < \varepsilon.$$

($\Leftarrow$) Suppose (a_n) is Cauchy. First, (a_n) is bounded: choose N so that $|a_n - a_m| < 1$ for $n, m \geq N$; then $|a_n| \leq \max(|a_1|, \ldots, |a_{N-1}|, |a_N| + 1)$ for all n.

By Bolzano–Weierstrass (Theorem 3.4), (a_n) has a convergent subsequence $a_{n_k} \to L$. We show $a_n \to L$. Given $\varepsilon > 0$, choose N_1 so that $|a_n - a_m| < \varepsilon/2$ for $n, m \geq N_1$, and choose K so that $|a_{n_k} - L| < \varepsilon/2$ for $k \geq K$. Pick $k \geq K$ with $n_k \geq N_1$. Then for $n \geq N_1$:

$$|a_n - L| \leq |a_n - a_{n_k}| + |a_{n_k} - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$ □

3.4 Limit Superior and Limit Inferior

Definition 3.3 (lim sup and lim inf)**.** For a bounded sequence (a_n), define:

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty}\Big(\sup_{k\geq n} a_k\Big), \qquad \liminf_{n\to\infty} a_n = \lim_{n\to\infty}\Big(\inf_{k\geq n} a_k\Big).$$

The sequences $\sup_{k\geq n} a_k$ (decreasing) and $\inf_{k\geq n} a_k$ (increasing) are monotone and bounded, so both limits exist by Theorem 3.3.

Proposition 3.6 (Properties of lim sup and lim inf)**.** *Let (a_n) be a bounded sequence. Then:*

(i) $\liminf_{n\to\infty} a_n \leq \limsup_{n\to\infty} a_n$.

(ii) *(a_n) converges if and only if $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n$, and in that case the limit equals this common value.*

(iii) *$\limsup_{n\to\infty} a_n = L$ if and only if: for every $\varepsilon > 0$, (a) $a_n < L + \varepsilon$ for all sufficiently large n, and (b) $a_n > L - \varepsilon$ for infinitely many n.*

Proof. (i) For each n, $\inf_{k\geq n} a_k \leq \sup_{k\geq n} a_k$. Taking limits preserves the inequality.

(ii) ($\Rightarrow$) Suppose $a_n \to L$. Given $\varepsilon > 0$, for large n we have $L - \varepsilon < a_k < L + \varepsilon$ for all $k \geq n$, so $L - \varepsilon \leq \inf_{k\geq n} a_k \leq \sup_{k\geq n} a_k \leq L + \varepsilon$. Taking the limit gives $L - \varepsilon \leq \liminf a_n \leq \limsup a_n \leq L + \varepsilon$ for all $\varepsilon > 0$, so both equal L.

($\Leftarrow$) Suppose $\liminf a_n = \limsup a_n = L$. For any $\varepsilon > 0$, choose N so that $|\sup_{k\geq n} a_k - L| < \varepsilon$ and $|\inf_{k\geq n} a_k - L| < \varepsilon$ for $n \geq N$. Then for $n \geq N$:

$$L - \varepsilon < \inf_{k\geq n} a_k \leq a_n \leq \sup_{k\geq n} a_k < L + \varepsilon,$$

so $|a_n - L| < \varepsilon$.

(iii) Let $s_n = \sup_{k\geq n} a_k$ and $L = \lim s_n$. For (a): given $\varepsilon > 0$, choose N with $s_N < L + \varepsilon$. Then $a_n \leq s_N < L + \varepsilon$ for all $n \geq N$. For (b): given $\varepsilon > 0$, for every n we have $s_n \geq L$ (since s_n is decreasing to L), so there exists $k \geq n$ with $a_k > s_n - \varepsilon \geq L - \varepsilon$. Since n is arbitrary, this gives infinitely many such k.

Conversely, suppose (a) and (b) hold for some value L'. From (a), $s_n \leq L' + \varepsilon$ for large n, so $\limsup a_n \leq L' + \varepsilon$. From (b), for each n there exists $k \geq n$ with $a_k > L' - \varepsilon$, so $s_n \geq L' - \varepsilon$, giving $\limsup a_n \geq L' - \varepsilon$. Since ε is arbitrary, $\limsup a_n = L'$. □

3.5 Series

Definition 3.4 (Convergence of Series)**.** Let $(a_n)_{n=1}^{\infty}$ be a sequence. The series $\sum_{n=1}^{\infty} a_n$ *converges* if the sequence of partial sums $S_N = \sum_{n=1}^{N} a_n$ converges. The series *converges absolutely* if $\sum_{n=1}^{\infty} |a_n|$ converges.

Theorem 3.7 (Comparison Test)**.** *If $0 \leq a_n \leq b_n$ for all n and $\sum b_n$ converges, then $\sum a_n$ converges.*

Proof. The partial sums $S_N = \sum_{n=1}^{N} a_n$ form an increasing sequence (since $a_n \geq 0$) bounded above by $\sum_{n=1}^{\infty} b_n$. By the monotone convergence theorem (Theorem 3.3), (S_N) converges. □

Theorem 3.8 (Absolute Convergence Implies Convergence)**.** *If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.*

Proof. Absolute convergence means the partial sums of $|a_n|$ form a Cauchy sequence. For $m > n$:

$$\Big|\sum_{k=n+1}^{m} a_k\Big| \leq \sum_{k=n+1}^{m} |a_k|,$$

so the partial sums of a_n are also Cauchy, hence convergent by Theorem 3.5. □

Theorem 3.9 (Ratio Test)**.** *Let (a_n) be a sequence with $a_n \neq 0$ for all n. If $\limsup_{n\to\infty} |a_{n+1}/a_n| < 1$, then $\sum a_n$ converges absolutely. If $\liminf_{n\to\infty} |a_{n+1}/a_n| > 1$, then $\sum a_n$ diverges.*

Proof. Suppose $\limsup |a_{n+1}/a_n| = L < 1$. Choose r with $L < r < 1$. By Proposition 3.6(iii), there exists N with $|a_{n+1}/a_n| < r$ for all $n \geq N$. Then $|a_n| \leq |a_N| r^{n-N}$ for $n \geq N$, and $\sum |a_N| r^{n-N}$ converges (geometric series with ratio $r < 1$). By the comparison test, $\sum |a_n|$ converges.

If $\liminf |a_{n+1}/a_n| > 1$, then $|a_{n+1}| > |a_n|$ for large n, so $a_n \not\to 0$ and the series diverges. □

Theorem 3.10 (Root Test)**.** *Let $\alpha = \limsup_{n\to\infty} |a_n|^{1/n}$. If $\alpha < 1$, then $\sum a_n$ converges absolutely. If $\alpha > 1$, then $\sum a_n$ diverges.*

Proof. If $\alpha < 1$, choose r with $\alpha < r < 1$. By Proposition 3.6(iii), $|a_n|^{1/n} < r$ for all sufficiently large n, so $|a_n| < r^n$. The comparison test with the geometric series $\sum r^n$ gives absolute convergence.

If $\alpha > 1$, then $|a_n|^{1/n} > 1$ for infinitely many n (Proposition 3.6(iii)), so $|a_n| > 1$ infinitely often, meaning $a_n \not\to 0$ and the series diverges. □

Remark 3.1 (Conditional Convergence). A series that converges but does not converge absolutely is called *conditionally convergent*. The classic example is the alternating harmonic series $\sum(-1)^{n+1}/n$. By the Riemann rearrangement theorem, any conditionally convergent series can be rearranged to converge to any prescribed value (or to diverge). This pathology motivates the focus on absolute convergence and non-negative series in measure theory.

4 Topology of the Real Line

4.1 Open and Closed Sets

Definition 4.1 (Open and Closed Sets)**.** A set $U \subseteq \mathbb{R}$ is *open* if for every $x \in U$ there exists $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \subseteq U$. A set $F \subseteq \mathbb{R}$ is *closed* if its complement $\mathbb{R} \setminus F$ is open.

Proposition 4.1. *(i) The empty set $\emptyset$ and $\mathbb{R}$ are both open and closed.*
(ii) An arbitrary union of open sets is open.
(iii) A finite intersection of open sets is open.
(iv) An arbitrary intersection of closed sets is closed.
(v) A finite union of closed sets is closed.

Proof. (i) $\emptyset$ is vacuously open; $\mathbb{R}$ is open since every ε-neighborhood is contained in $\mathbb{R}$. Their complements give the closed assertions.

(ii) If $x \in \bigcup_\alpha U_\alpha$ with each U_α open, then $x \in U_\beta$ for some β, so some ε-neighborhood of x is contained in $U_\beta \subseteq \bigcup_\alpha U_\alpha$.

(iii) If $x \in \bigcap_{i=1}^n U_i$ with each U_i open, choose $\varepsilon_i > 0$ with $(x - \varepsilon_i, x + \varepsilon_i) \subseteq U_i$. Then $\varepsilon = \min(\varepsilon_1, \ldots, \varepsilon_n) > 0$ and $(x - \varepsilon, x + \varepsilon) \subseteq \bigcap_{i=1}^n U_i$.

(iv) and (v) follow from (ii) and (iii) by taking complements via De Morgan's laws. □

Proposition 4.2 (Sequential Characterization of Closed Sets)**.** *A set $F \subseteq \mathbb{R}$ is closed if and only if it contains all its limit points: whenever (x_n) is a sequence in F with $x_n \to L$, we have $L \in F$.*

Proof. ($\Rightarrow$) Suppose F is closed, $(x_n) \subseteq F$, and $x_n \to L$. If $L \notin F$, then $L \in \mathbb{R} \setminus F$ which is open, so there exists $\varepsilon > 0$ with $(L - \varepsilon, L + \varepsilon) \subseteq \mathbb{R} \setminus F$. But $x_n \to L$ gives $x_n \in (L - \varepsilon, L + \varepsilon)$ for large n, contradicting $x_n \in F$.

($\Leftarrow$) Suppose F contains all its limit points. We show $\mathbb{R} \setminus F$ is open. Let $x \in \mathbb{R} \setminus F$. If no ε-neighborhood of x is contained in $\mathbb{R} \setminus F$, then for each n there exists $x_n \in F$ with $|x_n - x| < 1/n$, giving $x_n \to x$ with $(x_n) \subseteq F$, so $x \in F$, a contradiction. □

Definition 4.2 (Closure, Interior, Boundary)**.** Let $S \subseteq \mathbb{R}$.
(i) The *closure* $\overline{S}$ is the smallest closed set containing S: $\overline{S} = \bigcap\{F : F \text{ closed}, S \subseteq F\}$. Equivalently, $x \in \overline{S}$ iff every neighborhood of x intersects S.
(ii) The *interior* S° is the largest open set contained in S: $S^\circ = \bigcup\{U : U \text{ open}, U \subseteq S\}$.
(iii) The *boundary* $\partial S = \overline{S} \setminus S^\circ$.

Example 4.1. $\overline{\mathbb{Q}} = \mathbb{R}$ and $\mathbb{Q}^\circ = \emptyset$ (by density of both rationals and irrationals). Thus $\partial\mathbb{Q} = \mathbb{R}$.

4.2 Compactness and the Heine–Borel Theorem

Definition 4.3 (Open Cover, Compactness)**.** An *open cover* of $S \subseteq \mathbb{R}$ is a collection $\{U_\alpha\}$ of open sets with $S \subseteq \bigcup_\alpha U_\alpha$. A set $K \subseteq \mathbb{R}$ is *compact* if every open cover of K has a finite subcover.

Theorem 4.3 (Heine–Borel)**.** *A subset $K \subseteq \mathbb{R}$ is compact if and only if it is closed and bounded.*

Proof. ($\Rightarrow$) Suppose K is compact.

K is bounded: The collection $\{(-n, n) : n \in \mathbb{N}\}$ is an open cover of K. By compactness, finitely many of these cover K, say $K \subseteq (-N, N)$. So K is bounded.

K is closed: Let $x \notin K$. For each $y \in K$, let $r_y = |x - y|/2 > 0$. The open sets $U_y = (y - r_y, y + r_y)$ cover K. By compactness, $K \subseteq U_{y_1} \cup \cdots \cup U_{y_m}$ for some finite collection. Let $\delta = \min(r_{y_1}, \ldots, r_{y_m}) > 0$. Then $(x-\delta, x+\delta) \cap K = \emptyset$ (since any z in this neighborhood satisfies $|z - x| < \delta \leq r_{y_i}$, hence $|z - y_i| \geq |x - y_i| - |x - z| > 2r_{y_i} - r_{y_i} = r_{y_i}$, so $z \notin U_{y_i}$ for each i). Thus $\mathbb{R} \setminus K$ is open, so K is closed.

($\Leftarrow$) Suppose K is closed and bounded, say $K \subseteq [a, b]$. Let $\{U_\alpha\}$ be an open cover of K. We show it has a finite subcover by contradiction.

Bisect $[a, b]$ into $[a, (a + b)/2]$ and $[(a + b)/2, b]$. At least one half, call it $[a_1, b_1]$, has the property that $K \cap [a_1, b_1]$ is not covered by any finite subcollection of $\{U_\alpha\}$. Continue bisecting: at step n, we have $[a_n, b_n]$ with $b_n - a_n = (b - a)/2^n$ such that $K \cap [a_n, b_n]$ cannot be finitely covered.

By the nested intervals property (Theorem 2.5), there exists $x \in \bigcap_{n=1}^{\infty} [a_n, b_n]$. Since each $[a_n, b_n]$ intersects K nontrivially (otherwise a finite — in fact, empty — subcover would exist), choose $x_n \in K \cap [a_n, b_n]$. Then $|x_n - x| \leq b_n - a_n = (b - a)/2^n \to 0$, so $x_n \to x$. Since K is closed, $x \in K$.

Since $\{U_\alpha\}$ covers K, there exists U_β with $x \in U_\beta$. Since U_β is open, there exists $\varepsilon > 0$ with $(x-\varepsilon, x+\varepsilon) \subseteq U_\beta$. For n large enough that $(b-a)/2^n < \varepsilon$, we have $[a_n, b_n] \subseteq (x-\varepsilon, x+\varepsilon) \subseteq U_\beta$. Then $\{U_\beta\}$ is a finite subcover of $K \cap [a_n, b_n]$, a contradiction. □

Proposition 4.4 (Properties of Compact Sets). *(i) A closed subset of a compact set is compact.*

(ii) A compact set is sequentially compact: every sequence in K has a subsequence converging to a point in K.

Proof. (i) Let $F \subseteq K$ be closed with K compact. If $\{U_\alpha\}$ is an open cover of F, then $\{U_\alpha\} \cup \{\mathbb{R} \setminus F\}$ is an open cover of K (since $\mathbb{R} \setminus F$ is open). By compactness of K, a finite subcover exists. Removing $\mathbb{R} \setminus F$ (if present) leaves a finite subcover of F.

(ii) If (x_n) is a sequence in K, then (x_n) is bounded (since K is bounded by Heine–Borel). By Bolzano–Weierstrass, a subsequence converges to some L. Since K is closed, $L \in K$. □

Compactness Beyond $\mathbb{R}^n$

Compactness is used throughout measure theory and functional analysis. The Heine–Borel theorem characterizes compact subsets of $\mathbb{R}^n$. In functional analysis (Phase 01), compactness in infinite-dimensional spaces is more subtle and requires the Arzelà–Ascoli theorem.

5 Continuity

5.1 Definition and Sequential Characterization

Definition 5.1 (Continuity). Let $f : A \to \mathbb{R}$ with $A \subseteq \mathbb{R}$ and let $c \in A$. We say f is *continuous at c* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|f(x) - f(c)| < \varepsilon$ whenever $x \in A$ and $|x - c| < \delta$. We say f is *continuous on A* if it is continuous at every $c \in A$.

Theorem 5.1 (Sequential Characterization of Continuity). *Let $f : A \to \mathbb{R}$ and $c \in A$. Then f is continuous at c if and only if for every sequence (x_n) in A with $x_n \to c$, we have $f(x_n) \to f(c)$.*

Proof. ($\Rightarrow$) Suppose f is continuous at c and $x_n \to c$. Given $\varepsilon > 0$, choose $\delta > 0$ from the definition of continuity, then choose N with $|x_n - c| < \delta$ for $n \geq N$. Then $|f(x_n) - f(c)| < \varepsilon$ for $n \geq N$.

($\Leftarrow$) We prove the contrapositive. Suppose f is not continuous at c. Then there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$ there exists $x \in A$ with $|x - c| < \delta$ but $|f(x) - f(c)| \geq \varepsilon_0$. Taking $\delta = 1/n$, we obtain a sequence $x_n \in A$ with $|x_n - c| < 1/n$ (so $x_n \to c$) but $|f(x_n) - f(c)| \geq \varepsilon_0$ (so $f(x_n) \not\to f(c)$). □

Proposition 5.2 (Algebraic Properties of Continuous Functions)**.** *If $f, g : A \to \mathbb{R}$ are continuous at $c \in A$, then so are $f + g$, fg, and f/g (the last provided $g(c) \neq 0$). If $g : A \to \mathbb{R}$ is continuous at c and $h : B \to \mathbb{R}$ is continuous at $g(c)$ with $g(A) \subseteq B$, then $h \circ g$ is continuous at c.*

Proof. The first three assertions follow immediately from Theorem 5.1 and the algebraic limit theorems (Theorem 3.2). For composition: if $x_n \to c$, then $g(x_n) \to g(c)$ by continuity of g, then $h(g(x_n)) \to h(g(c))$ by continuity of h. □

5.2 Extreme Value Theorem and Intermediate Value Theorem

Theorem 5.3 (Extreme Value Theorem)**.** *If $f : K \to \mathbb{R}$ is continuous and K is compact, then f attains its maximum and minimum: there exist $x_*, x^* \in K$ with $f(x_*) \leq f(x) \leq f(x^*)$ for all $x \in K$.*

Proof. We show f attains its maximum; the minimum argument is analogous (apply the result to $-f$).

Step 1: f is bounded above. Suppose not. Then for each n there exists $x_n \in K$ with $f(x_n) > n$. By sequential compactness (Proposition 4.4(ii)), (x_n) has a subsequence $x_{n_k} \to c \in K$. By continuity, $f(x_{n_k}) \to f(c)$, but $f(x_{n_k}) > n_k \to \infty$, a contradiction.

Step 2: The supremum is attained. Let $M = \sup\{f(x) : x \in K\}$, which is finite by Step 1. By the characterization of supremum, for each n there exists $x_n \in K$ with $f(x_n) > M - 1/n$. By sequential compactness, $x_{n_k} \to x^* \in K$ for some subsequence. By continuity, $f(x^*) = \lim f(x_{n_k}) = M$. □

Theorem 5.4 (Intermediate Value Theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) < v < f(b)$ (or $f(a) > v > f(b)$), then there exists $c \in (a, b)$ with $f(c) = v$.*

Proof. Assume $f(a) < v < f(b)$. Let $S = \{x \in [a, b] : f(x) < v\}$. Then S is nonempty (since $a \in S$) and bounded above (by b), so $c = \sup S$ exists with $a \leq c \leq b$.

Claim: $f(c) = v$. Choose a sequence (s_n) in S with $s_n \to c$. By continuity, $f(c) = \lim f(s_n) \leq v$ (since $f(s_n) < v$). Now if $f(c) < v$, then by continuity there exists $\delta > 0$ with $f(x) < v$ for $x \in (c - \delta, c + \delta) \cap [a, b]$. Since $c < b$ (because $f(c) < v < f(b)$), there exists $x \in (c, c + \delta) \cap [a, b]$ with $f(x) < v$, contradicting $c = \sup S$. Therefore $f(c) = v$. □

5.3 Uniform Continuity

Definition 5.2 (Uniform Continuity)**.** A function $f : A \to \mathbb{R}$ is *uniformly continuous* on A if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ whenever $x, y \in A$ with $|x - y| < \delta$.

Remark 5.1. The distinction from ordinary continuity: in uniform continuity, δ depends only on ε, not on the point. Ordinary continuity allows δ to vary with the point c.

Theorem 5.5 (Continuous on Compact $\Rightarrow$ Uniformly Continuous)**.** *If $K \subseteq \mathbb{R}$ is compact and $f : K \to \mathbb{R}$ is continuous, then f is uniformly continuous on K.*

Proof. Suppose f is not uniformly continuous on K. Then there exists $\varepsilon_0 > 0$ such that for every $n \in \mathbb{N}$ there exist $x_n, y_n \in K$ with $|x_n - y_n| < 1/n$ but $|f(x_n) - f(y_n)| \geq \varepsilon_0$.

By sequential compactness, (x_n) has a subsequence $x_{n_k} \to c \in K$. Since $|y_{n_k} - c| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - c| < 1/n_k + |x_{n_k} - c| \to 0$, we also have $y_{n_k} \to c$. By continuity of f at c:

$$|f(x_{n_k}) - f(y_{n_k})| \leq |f(x_{n_k}) - f(c)| + |f(c) - f(y_{n_k})| \to 0,$$

contradicting $|f(x_{n_k}) - f(y_{n_k})| \geq \varepsilon_0$. □

6 Differentiation

6.1 Definition and Basic Properties

Definition 6.1 (Derivative)**.** Let $f : (a, b) \to \mathbb{R}$ and $c \in (a, b)$. The *derivative of f at c* is

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h},$$

provided this limit exists. If $f'(c)$ exists, we say f is *differentiable at c*.

Theorem 6.1 (Differentiable Implies Continuous)**.** *If f is differentiable at c, then f is continuous at c.*

Proof. We have

$$f(c+h) - f(c) = \frac{f(c+h) - f(c)}{h} \cdot h \to f'(c) \cdot 0 = 0$$

as $h \to 0$, so $f(c+h) \to f(c)$. □

Proposition 6.2 (Algebraic Differentiation Rules)**.** *If f and g are differentiable at c, then:*

(i) $(f+g)'(c) = f'(c) + g'(c)$.
(ii) (Product rule) $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$.
(iii) (Quotient rule) If $g(c) \neq 0$, then $(f/g)'(c) = \big(f'(c)g(c) - f(c)g'(c)\big)/g(c)^2$.

Proof. We prove (ii). Write

$$\begin{aligned} f(c+h)g(c+h) - f(c)g(c) &= f(c+h)g(c+h) - f(c)g(c+h) + f(c)g(c+h) - f(c)g(c) \\ &= \big(f(c+h) - f(c)\big)g(c+h) + f(c)\big(g(c+h) - g(c)\big). \end{aligned}$$

Dividing by h and taking $h \to 0$, using $g(c+h) \to g(c)$ (since g is continuous at c by Theorem 6.1), gives $f'(c)g(c) + f(c)g'(c)$. □

Theorem 6.3 (Chain Rule)**.** *Let g be differentiable at c and f be differentiable at $g(c)$. Then $f \circ g$ is differentiable at c and $(f \circ g)'(c) = f'(g(c)) \cdot g'(c)$.*

Proof. Define the auxiliary function

$$\varphi(y) = \begin{cases} \dfrac{f(y) - f(g(c))}{y - g(c)} & \text{if } y \neq g(c), \\ f'(g(c)) & \text{if } y = g(c). \end{cases}$$

Since f is differentiable at $g(c)$, φ is continuous at $g(c)$. For $h \neq 0$ with $g(c+h) \neq g(c)$:

$$\frac{f(g(c+h)) - f(g(c))}{h} = \varphi(g(c+h)) \cdot \frac{g(c+h) - g(c)}{h}.$$

This identity also holds when $g(c+h) = g(c)$ (both sides are zero, using the convention that $\varphi(g(c)) = f'(g(c))$ and $g(c+h) - g(c) = 0$). Taking $h \to 0$: $\varphi(g(c+h)) \to \varphi(g(c)) = f'(g(c))$ (by continuity of φ at $g(c)$ and continuity of g at c), and $(g(c+h) - g(c))/h \to g'(c)$. Therefore $(f \circ g)'(c) = f'(g(c)) \cdot g'(c)$. □

6.2 Mean Value Theorem and Consequences

Theorem 6.4 (Fermat's Theorem on Local Extrema)**.** *If $f : (a, b) \to \mathbb{R}$ is differentiable at $c \in (a, b)$ and f has a local extremum at c, then $f'(c) = 0$.*

Proof. Suppose f has a local maximum at c (the minimum case is analogous). There exists $\delta > 0$ with $f(x) \leq f(c)$ for $|x - c| < \delta$. For $0 < h < \delta$:

$$\frac{f(c+h) - f(c)}{h} \leq 0,$$

so $f'(c) = \lim_{h\to 0^+} \frac{f(c+h)-f(c)}{h} \leq 0$. For $-\delta < h < 0$:

$$\frac{f(c+h) - f(c)}{h} \geq 0,$$

so $f'(c) = \lim_{h\to 0^-} \frac{f(c+h)-f(c)}{h} \geq 0$. Together, $f'(c) = 0$. □

Theorem 6.5 (Rolle's Theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then there exists $c \in (a, b)$ with $f'(c) = 0$.*

Proof. If f is constant on $[a, b]$, then $f'(c) = 0$ for all $c \in (a, b)$. Otherwise, f is not identically $f(a)$, so f attains a maximum or minimum value different from $f(a) = f(b)$. By the extreme value theorem (Theorem 5.3), this extremum is attained at some $c \in [a, b]$. Since the extreme value differs from $f(a) = f(b)$, the point c cannot be an endpoint, so $c \in (a, b)$. By Fermat's theorem (Theorem 6.4), $f'(c) = 0$. □

Theorem 6.6 (Mean Value Theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ with*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Define $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x - a)$. Then g is continuous on $[a, b]$, differentiable on (a, b), and $g(a) = f(a) = g(b)$ (verify: $g(b) = f(b) - (f(b) - f(a)) = f(a)$). By Rolle's theorem, there exists $c \in (a, b)$ with $0 = g'(c) = f'(c) - \frac{f(b)-f(a)}{b-a}$. □

6.3 Taylor's Theorem

Theorem 6.7 (Taylor's Theorem with Lagrange Remainder)**.** *Let $f : [a, b] \to \mathbb{R}$ have continuous derivatives $f', f'', \ldots, f^{(n)}$ on $[a, b]$ and let $f^{(n+1)}$ exist on (a, b). Then for any $x \in [a, b]$ and $c \in [a, b]$:*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x - c)^k + \frac{f^{(n+1)}(\xi)}{(n + 1)!}(x - c)^{n+1}$$

for some ξ between c and x.

Proof. We may assume $x \neq c$; the result is trivial otherwise. Define the remainder $R = f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x - c)^k$ and choose M so that

$$R = \frac{M}{(n + 1)!}(x - c)^{n+1}.$$

We need to show $M = f^{(n+1)}(\xi)$ for some ξ between c and x. Define

$$g(t) = f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x - t)^k - \frac{M}{(n + 1)!}(x - t)^{n+1}.$$

Then $g(c) = f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k - \frac{M}{(n+1)!}(x-c)^{n+1} = R - R = 0$, and $g(x) = f(x) - f(x) = 0$. By Rolle's theorem, there exists ξ between c and x with $g'(\xi) = 0$.

Computing $g'(t)$ by differentiating each term and using telescoping cancellation:

$$g'(t) = -\sum_{k=0}^{n} \left[\frac{f^{(k+1)}(t)}{k!}(x-t)^k - \frac{f^{(k)}(t)}{(k-1)!}(x-t)^{k-1} \right] + \frac{M}{n!}(x-t)^n,$$

where the sum telescopes (the second part of the k-th term cancels the first part of the $(k-1)$-th term), leaving

$$g'(t) = -\frac{f^{(n+1)}(t)}{n!}(x-t)^n + \frac{M}{n!}(x-t)^n = \frac{(x-t)^n}{n!}\big(M - f^{(n+1)}(t)\big).$$

Setting $g'(\xi) = 0$ and noting $(x-\xi)^n \neq 0$ (since ξ is between c and x, and $c \neq x$) gives $M = f^{(n+1)}(\xi)$. □

Taylor's Theorem in Optimization and Measure Theory

Taylor's theorem is essential for optimization theory (Phase 01). Gradient descent convergence proofs use the first-order Taylor expansion with remainder to bound function value changes. The mean value theorem appears in many ε-δ arguments throughout measure theory.

7 Riemann Integration

7.1 Partitions and Riemann Sums

Definition 7.1 (Partition)**.** A *partition* of $[a, b]$ is a finite set $P = \{x_0, x_1, \ldots, x_n\}$ with $a = x_0 < x_1 < \cdots < x_n = b$.

Definition 7.2 (Upper and Lower Sums)**.** Let $f : [a, b] \to \mathbb{R}$ be bounded and $P = \{x_0, \ldots, x_n\}$ a partition. Define

$$M_k = \sup_{x \in [x_{k-1}, x_k]} f(x), \qquad m_k = \inf_{x \in [x_{k-1}, x_k]} f(x), \qquad \Delta x_k = x_k - x_{k-1}.$$

The *upper sum* and *lower sum* are

$$U(f, P) = \sum_{k=1}^{n} M_k \, \Delta x_k, \qquad L(f, P) = \sum_{k=1}^{n} m_k \, \Delta x_k.$$

Lemma 7.1 (Refinement Lemma)**.** *If Q is a refinement of P (i.e., $P \subseteq Q$), then $L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P)$.*

Proof. It suffices to show this when Q has one more point than P. Say $Q = P \cup \{x^*\}$ with $x_{k-1} < x^* < x_k$. On $[x_{k-1}, x_k]$, the lower sum contribution changes from $m_k(x_k - x_{k-1})$ to $m'(x^* - x_{k-1}) + m''(x_k - x^*)$ where $m' = \inf_{[x_{k-1}, x^*]} f \geq m_k$ and $m'' = \inf_{[x^*, x_k]} f \geq m_k$. Thus the lower sum does not decrease. The upper sum argument is analogous (infima become suprema, inequality reverses). The middle inequality $L(f, Q) \leq U(f, Q)$ is immediate from $m_k \leq M_k$. □

Corollary 7.2. *For any two partitions P, Q: $L(f, P) \leq U(f, Q)$.*

Proof. $L(f, P) \leq L(f, P \cup Q) \leq U(f, P \cup Q) \leq U(f, Q)$. □

Definition 7.3 (Riemann Integrability)**.** A bounded function $f : [a, b] \to \mathbb{R}$ is *Riemann integrable* if

$$\sup_P L(f, P) = \inf_P U(f, P),$$

where the supremum and infimum are over all partitions P of $[a, b]$. This common value is the *Riemann integral* $\int_a^b f(x)\,dx$. By Corollary 7.2, the sup and inf exist and satisfy $\sup_P L(f, P) \leq \inf_P U(f, P)$.

Remark 7.1 (Riemann Criterion). Equivalently, f is Riemann integrable if and only if for every $\varepsilon > 0$ there exists a partition P with $U(f, P) - L(f, P) < \varepsilon$.

7.2 Continuous Functions Are Riemann Integrable

Theorem 7.3. *If $f : [a, b] \to \mathbb{R}$ is continuous, then f is Riemann integrable.*

Proof. Since $[a, b]$ is compact, f is uniformly continuous (Theorem 5.5). Given $\varepsilon > 0$, choose $\delta > 0$ with $|f(x) - f(y)| < \varepsilon/(b - a)$ whenever $|x - y| < \delta$. Let P be any partition with mesh $\|P\| = \max_k \Delta x_k < \delta$. On each subinterval $[x_{k-1}, x_k]$, continuity on the compact interval gives $M_k - m_k = f(u_k) - f(v_k)$ for some $u_k, v_k \in [x_{k-1}, x_k]$ (extreme value theorem), and $|u_k - v_k| \leq \Delta x_k < \delta$, so $M_k - m_k < \varepsilon/(b - a)$. Therefore:

$$U(f, P) - L(f, P) = \sum_{k=1}^{n} (M_k - m_k)\Delta x_k < \frac{\varepsilon}{b - a} \sum_{k=1}^{n} \Delta x_k = \varepsilon.$$

By the Riemann criterion (Remark 7.1), f is integrable. □

7.3 Properties of the Riemann Integral

Proposition 7.4 (Properties)**.** *Let $f, g : [a, b] \to \mathbb{R}$ be Riemann integrable. Then:*

(i) ***Linearity:*** *$\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g$ for all $\alpha, \beta \in \mathbb{R}$.*

(ii) ***Monotonicity:*** *If $f(x) \leq g(x)$ for all $x \in [a, b]$, then $\int_a^b f \leq \int_a^b g$.*

(iii) ***Additivity:*** *For $a < c < b$, f is integrable on $[a, c]$ and $[c, b]$, and $\int_a^b f = \int_a^c f + \int_c^b f$.*

(iv) ***Triangle inequality:*** *$|f|$ is integrable and $|\int_a^b f| \leq \int_a^b |f|$.*

Proof. We prove (ii). For any partition P, $\inf_{[x_{k-1},x_k]} f \leq \inf_{[x_{k-1},x_k]} g$ is *not* generally true; rather we use: $\sup_P L(f, P) \leq \sup_P L(g, P)$ since for each partition, $m_k^f \leq m_k^g$ implies $L(f, P) \leq L(g, P)$. Wait — this is incorrect since $m_k^f \leq m_k^g$ does *not* follow from $f \leq g$. Actually it does: if $f(x) \leq g(x)$ for all x, then $\inf_{[x_{k-1},x_k]} f(x) \leq \inf_{[x_{k-1},x_k]} g(x)$. Therefore $L(f, P) \leq L(g, P)$ for all P, and $\int_a^b f = \sup_P L(f, P) \leq \sup_P L(g, P) = \int_a^b g$.

We prove (i) for the case $\alpha = 1, \beta = 1$ (the general case follows similarly). Given $\varepsilon > 0$, choose partitions P_1, P_2 with $U(f, P_1) - L(f, P_1) < \varepsilon/2$ and $U(g, P_2) - L(g, P_2) < \varepsilon/2$. Let $P = P_1 \cup P_2$. On each subinterval of P, $\sup(f + g) \leq \sup f + \sup g$ and $\inf(f + g) \geq \inf f + \inf g$. Therefore $U(f + g, P) \leq U(f, P) + U(g, P)$ and $L(f + g, P) \geq L(f, P) + L(g, P)$. Thus:

$$U(f + g, P) - L(f + g, P) \leq \big(U(f, P) - L(f, P)\big) + \big(U(g, P) - L(g, P)\big) < \varepsilon,$$

so $f + g$ is integrable. Moreover, $L(f, P) + L(g, P) \leq L(f + g, P) \leq \int_a^b (f + g) \leq U(f + g, P) \leq U(f, P) + U(g, P)$. Since $\int_a^b f + \int_a^b g$ also lies between $L(f, P) + L(g, P)$ and $U(f, P) + U(g, P)$, the two values differ by at most ε. As ε is arbitrary, equality holds. □

7.4 Fundamental Theorem of Calculus

Theorem 7.5 (Fundamental Theorem of Calculus, Part I)**.** *Let $f : [a, b] \to \mathbb{R}$ be Riemann integrable. Define $F(x) = \int_a^x f(t)\,dt$ for $x \in [a, b]$. Then F is continuous on $[a, b]$. If f is continuous at $c \in (a, b)$, then F is differentiable at c and $F'(c) = f(c)$.*

Proof. Continuity: Since f is bounded, say $|f(t)| \leq M$, for $x, y \in [a, b]$ with $x < y$:

$$|F(y) - F(x)| = \Big|\int_x^y f(t)\,dt\Big| \leq M(y - x).$$

So $|F(y) - F(x)| \leq M|y - x|$, and F is (Lipschitz) continuous.

Differentiability: Suppose f is continuous at c. Given $\varepsilon > 0$, choose $\delta > 0$ with $|f(t) - f(c)| < \varepsilon$ for $|t - c| < \delta$. For $0 < |h| < \delta$:

$$\frac{F(c+h) - F(c)}{h} - f(c) = \frac{1}{h}\int_c^{c+h} f(t)\,dt - f(c) = \frac{1}{h}\int_c^{c+h} \big(f(t) - f(c)\big)\,dt.$$

Therefore

$$\Big|\frac{F(c+h) - F(c)}{h} - f(c)\Big| \leq \frac{1}{|h|}\int_{\min(c,c+h)}^{\max(c,c+h)} |f(t) - f(c)|\,dt < \frac{1}{|h|} \cdot \varepsilon \cdot |h| = \varepsilon. \qquad \square$$

Theorem 7.6 (Fundamental Theorem of Calculus, Part II)**.** *If $f : [a, b] \to \mathbb{R}$ is Riemann integrable and $F : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), with $F'(x) = f(x)$ for all $x \in (a, b)$, then*

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Proof. Let $P = \{x_0, x_1, \ldots, x_n\}$ be any partition of $[a, b]$. By the mean value theorem (Theorem 6.6), on each subinterval there exists $c_k \in (x_{k-1}, x_k)$ with $F(x_k) - F(x_{k-1}) = F'(c_k)\Delta x_k = f(c_k)\Delta x_k$. Therefore:

$$F(b) - F(a) = \sum_{k=1}^{n} \big(F(x_k) - F(x_{k-1})\big) = \sum_{k=1}^{n} f(c_k)\Delta x_k.$$

Since $m_k \leq f(c_k) \leq M_k$, we get $L(f, P) \leq F(b) - F(a) \leq U(f, P)$ for every partition P. Since f is integrable, the common value $\int_a^b f$ is the unique number between all lower and upper sums, so $\int_a^b f = F(b) - F(a)$. $\square$

Limitations of the Riemann Integral

The Riemann integral cannot handle many natural limits. For example, let $q_1, q_2, q_3, \ldots$ be an enumeration of the rationals in $[0, 1]$ and define $f_n = \mathbf{1}_{\{q_1,\ldots,q_n\}}$ (the indicator of the first n rationals). Each f_n is Riemann integrable with $\int_0^1 f_n = 0$, but the pointwise limit is the Dirichlet function $\mathbf{1}_{\mathbb{Q}}$, which is **not** Riemann integrable. This failure motivates the Lebesgue integral in Phase 01.

8 Sequences and Series of Functions

8.1 Pointwise and Uniform Convergence

Definition 8.1 (Pointwise Convergence)**.** A sequence of functions $f_n : A \to \mathbb{R}$ *converges pointwise* to $f : A \to \mathbb{R}$ if for every $x \in A$, $\lim_{n\to\infty} f_n(x) = f(x)$. That is, for every $x \in A$ and every $\varepsilon > 0$, there exists $N = N(x, \varepsilon) \in \mathbb{N}$ such that $|f_n(x) - f(x)| < \varepsilon$ for all $n \geq N$.

Definition 8.2 (Uniform Convergence)**.** A sequence $f_n : A \to \mathbb{R}$ *converges uniformly* to $f : A \to \mathbb{R}$ if for every $\varepsilon > 0$ there exists $N = N(\varepsilon) \in \mathbb{N}$ such that $|f_n(x) - f(x)| < \varepsilon$ for all $n \geq N$ and all $x \in A$.

Remark 8.1. Uniform convergence is equivalent to $\sup_{x\in A} |f_n(x) - f(x)| \to 0$ as $n \to \infty$. The key difference from pointwise convergence is that N does not depend on x.

Example 8.1 (Pointwise but Not Uniform Convergence)**.** Let $f_n : [0, 1] \to \mathbb{R}$ be defined by $f_n(x) = x^n$. For $x \in [0, 1)$, $f_n(x) = x^n \to 0$; and $f_n(1) = 1$ for all n. So $f_n \to f$ pointwise, where $f(x) = 0$ for $x \in [0, 1)$ and $f(1) = 1$. Each f_n is continuous, but the limit f is discontinuous at $x = 1$. The convergence is not uniform: $\sup_{x\in[0,1]} |f_n(x) - f(x)| \geq f_n(1 - 1/n) = (1 - 1/n)^n \to 1/e \neq 0$.

8.2 Interchange Theorems

Theorem 8.1 (Uniform Limit of Continuous Functions Is Continuous)**.** *Let $f_n : A \to \mathbb{R}$ be continuous for each n, and suppose $f_n \to f$ uniformly on A. Then f is continuous on A.*

Proof. Fix $c \in A$ and $\varepsilon > 0$. By uniform convergence, choose N with $|f_N(x) - f(x)| < \varepsilon/3$ for all $x \in A$. By continuity of f_N at c, choose $\delta > 0$ with $|f_N(x) - f_N(c)| < \varepsilon/3$ for $|x - c| < \delta$. Then for $|x - c| < \delta$:

$$\begin{aligned} |f(x) - f(c)| &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(c)| + |f_N(c) - f(c)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

□

Theorem 8.2 (Interchange of Limit and Integral Under Uniform Convergence)**.** *Let $f_n : [a, b] \to \mathbb{R}$ be Riemann integrable for each n, and suppose $f_n \to f$ uniformly on $[a, b]$. Then f is Riemann integrable and*

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \int_a^b f_n(x)\,dx.$$

Proof. f is integrable: Given $\varepsilon > 0$, choose N with $|f_N(x) - f(x)| < \varepsilon/(4(b-a))$ for all x. Since f_N is integrable, choose a partition P with $U(f_N, P) - L(f_N, P) < \varepsilon/2$. On each subinterval, $\sup f \leq \sup f_N + \varepsilon/(4(b - a))$ and $\inf f \geq \inf f_N - \varepsilon/(4(b - a))$, so

$$\begin{aligned} U(f, P) - L(f, P) &\leq U(f_N, P) - L(f_N, P) + \frac{2\varepsilon}{4(b-a)} \cdot (b - a) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Interchange: For $n \geq N$ with $|f_n(x) - f(x)| < \varepsilon/(b - a)$:

$$\Big|\int_a^b f_n - \int_a^b f\Big| = \Big|\int_a^b (f_n - f)\Big| \leq \int_a^b |f_n - f| < \frac{\varepsilon}{b-a} \cdot (b - a) = \varepsilon.$$

□

Theorem 8.3 (Weierstrass M-Test)**.** *Let $f_n : A \to \mathbb{R}$ be functions and suppose there exist constants $M_n \geq 0$ with $|f_n(x)| \leq M_n$ for all $x \in A$ and $\sum_{n=1}^{\infty} M_n < \infty$. Then $\sum_{n=1}^{\infty} f_n$ converges uniformly (and absolutely) on A.*

Proof. For each x, $\sum |f_n(x)| \leq \sum M_n < \infty$, so $\sum f_n(x)$ converges absolutely. Let $S(x) = \sum_{n=1}^{\infty} f_n(x)$ and $S_N(x) = \sum_{n=1}^{N} f_n(x)$. Then for all x:

$$|S(x) - S_N(x)| = \Big|\sum_{n=N+1}^{\infty} f_n(x)\Big| \leq \sum_{n=N+1}^{\infty} |f_n(x)| \leq \sum_{n=N+1}^{\infty} M_n.$$

The tail $\sum_{n=N+1}^{\infty} M_n \to 0$ as $N \to \infty$ (since $\sum M_n$ converges), and the bound is independent of x, so the convergence is uniform. □

From Uniform to Dominated Convergence

The interchange theorems for uniform convergence are the starting point. Measure theory (Phase 01) develops the far more powerful Monotone Convergence Theorem and Dominated Convergence Theorem, which allow interchange of limit and integral under much weaker conditions — pointwise convergence suffices when combined with a dominating function.

9 Metric Spaces

9.1 Definitions and Examples

Definition 9.1 (Metric Space)**.** A *metric space* is a pair (X, d) where X is a nonempty set and $d : X \times X \to [0, \infty)$ satisfies:

(i) $d(x, y) = 0$ if and only if $x = y$ (identity of indiscernibles),
(ii) $d(x, y) = d(y, x)$ for all $x, y \in X$ (symmetry),
(iii) $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$ (triangle inequality).

Example 9.1 (Standard Examples)**.** (a) $(\mathbb{R}^n, d_2)$ where $d_2(x, y) = \left(\sum_{i=1}^n (x_i - y_i)^2\right)^{1/2}$ is the Euclidean metric.
(b) (X, d_{disc}) where $d_{\text{disc}}(x, y) = 0$ if $x = y$ and 1 if $x \neq y$ is the *discrete metric* on any set X.
(c) $\big(C[a, b],\, d_\infty\big)$ where $C[a, b]$ is the set of continuous functions $f : [a, b] \to \mathbb{R}$ and $d_\infty(f, g) = \sup_{x\in[a,b]} |f(x) - g(x)|$ is the *sup metric* (or uniform metric). This is well-defined because continuous functions on a compact set are bounded (Theorem 5.3).

9.2 Topology, Convergence, and Completeness

Definition 9.2 (Open Ball, Open Set, Closed Set)**.** In a metric space (X, d):

(i) The *open ball* of radius $r > 0$ centered at x is $B(x, r) = \{y \in X : d(x, y) < r\}$.
(ii) A set $U \subseteq X$ is *open* if for every $x \in U$ there exists $r > 0$ with $B(x, r) \subseteq U$.
(iii) A set $F \subseteq X$ is *closed* if $X \setminus F$ is open.

Definition 9.3 (Convergence in a Metric Space)**.** A sequence (x_n) in (X, d) *converges* to $x \in X$ if $d(x_n, x) \to 0$, i.e., for every $\varepsilon > 0$ there exists N with $d(x_n, x) < \varepsilon$ for all $n \geq N$.

Definition 9.4 (Cauchy Sequence and Completeness)**.** A sequence (x_n) in (X, d) is *Cauchy* if for every $\varepsilon > 0$ there exists N with $d(x_n, x_m) < \varepsilon$ for all $n, m \geq N$. A metric space is *complete* if every Cauchy sequence converges (to an element of X).

Example 9.2. $\mathbb{R}$ is complete (Theorem 3.5). $\mathbb{Q}$ with the usual metric is not complete: the sequence of decimal approximations to $\sqrt{2}$ is Cauchy in $\mathbb{Q}$ but does not converge in $\mathbb{Q}$. The space $\big(C[a, b], d_\infty\big)$ is complete: this is a consequence of Theorem 8.1, since a Cauchy sequence in the sup metric converges uniformly, and the uniform limit of continuous functions is continuous.

9.3 Compactness in Metric Spaces

Definition 9.5 (Sequential Compactness)**.** A metric space (X, d) (or a subset $K \subseteq X$) is *sequentially compact* if every sequence in K has a subsequence converging to a point in K.

Remark 9.1. In a metric space, sequential compactness is equivalent to compactness (every open cover has a finite subcover). The Heine–Borel theorem is a special case for $\mathbb{R}^n$: a subset of $\mathbb{R}^n$ is (sequentially) compact if and only if it is closed and bounded.

9.4 The Contraction Mapping Theorem

Definition 9.6 (Contraction)**.** Let (X, d) be a metric space. A map $T : X \to X$ is a *contraction* (or *contraction mapping*) if there exists a constant $\gamma \in [0, 1)$ such that

$$d(T(x), T(y)) \leq \gamma\, d(x, y) \qquad \text{for all } x, y \in X.$$

The constant γ is called the *contraction factor*.

Theorem 9.1 (Banach Fixed Point Theorem / Contraction Mapping Theorem)**.** *Let (X, d) be a complete metric space and $T : X \to X$ a contraction with factor $\gamma \in [0, 1)$. Then:*

(i) T has a unique fixed point $x^* \in X$ (i.e., $T(x^*) = x^*$).
(ii) For any $x_0 \in X$, the iterates $x_{n+1} = T(x_n)$ satisfy $x_n \to x^*$.
(iii) The rate of convergence satisfies $d(x_n, x^*) \leq \frac{\gamma^n}{1-\gamma} d(x_0, x_1)$.

Proof. Existence. Fix any $x_0 \in X$ and define $x_{n+1} = T(x_n)$. We show (x_n) is Cauchy. By the contraction property, $d(x_{n+1}, x_n) \leq \gamma\, d(x_n, x_{n-1})$, so by induction $d(x_{n+1}, x_n) \leq \gamma^n d(x_1, x_0)$. For $m > n$:

$$d(x_m, x_n) \leq \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \leq \sum_{k=n}^{m-1} \gamma^k d(x_1, x_0) \leq \frac{\gamma^n}{1-\gamma} d(x_1, x_0).$$

Since $\gamma^n \to 0$, for any $\varepsilon > 0$ we can choose N so that $\gamma^N d(x_1, x_0)/(1-\gamma) < \varepsilon$. Then $d(x_m, x_n) < \varepsilon$ for $m > n \geq N$, so (x_n) is Cauchy. Since X is complete, $x_n \to x^*$ for some $x^* \in X$.

We verify $T(x^*) = x^*$. By the triangle inequality and contraction:

$$d(x^*, T(x^*)) \leq d(x^*, x_{n+1}) + d(T(x_n), T(x^*)) \leq d(x^*, x_{n+1}) + \gamma\, d(x_n, x^*) \to 0.$$

So $d(x^*, T(x^*)) = 0$, giving $T(x^*) = x^*$.

Uniqueness. If $T(y^*) = y^*$ as well, then $d(x^*, y^*) = d(T(x^*), T(y^*)) \leq \gamma\, d(x^*, y^*)$. Since $\gamma < 1$, this forces $d(x^*, y^*) = 0$, i.e., $x^* = y^*$.

Rate of convergence. Taking $m \to \infty$ in the estimate $d(x_m, x_n) \leq \gamma^n d(x_1, x_0)/(1-\gamma)$ gives $d(x^*, x_n) \leq \gamma^n d(x_1, x_0)/(1-\gamma)$. □

The Contraction Mapping Theorem in Reinforcement Learning

This theorem is fundamental to reinforcement learning. The Bellman optimality operator T defined by

$$(TV)(s) = \max_a \Big[R(s, a) + \gamma \sum_{s'} P(s' \mid s, a)\, V(s') \Big]$$

is a γ-contraction in the sup-norm on the space of bounded value functions. Value iteration — the algorithm that repeatedly applies T starting from any initial V_0 — converges to the unique fixed point V^* (the optimal value function) precisely because of the Banach fixed point theorem. The convergence rate $\gamma^n/(1-\gamma)$ gives the iteration complexity of value iteration.

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